



An orthogonal basis approach to formation shape control[☆]

Tairan Liu^a, Marcio de Queiroz^{b,*}

^a School of Electrical and Computer Engineering, University of Georgia, Athens, GA 30602, USA

^b Department of Mechanical and Industrial Engineering, Louisiana State University, Baton Rouge, LA 70803, USA

ARTICLE INFO

Article history:

Received 30 April 2020

Received in revised form 14 December 2020

Accepted 1 February 2021

Available online 15 April 2021

Keywords:

Formation control

Orthogonal basis

Distance

Signed volume

ABSTRACT

In this paper, we propose a novel approach to the problem of augmenting distance-based formation controllers with a secondary constraint for the purpose of preventing 3D formation ambiguities. Specifically, we introduce three controlled variables that form an orthogonal space and uniquely characterize a tetrahedron formation in 3D. This orthogonal space incorporates constraints on the inter-agent distances and the signed volume of tetrahedron substructures. The formation is modeled using a directed graph with a leader–follower type configuration and single-integrator dynamics. We show that the proposed decentralized formation controller ensures the *global* asymptotic stability and the local exponential stability of the desired formation for an n -agent system with no ambiguities. Unlike previous work, this result is achieved without conditions on the tetrahedrons that form the desired formation or on the control gains.

© 2021 Elsevier Ltd. All rights reserved.

1. Introduction

In formation shape control, a group of interacting mobile agents are commanded to acquire and maintain a desired geometric pattern in space. A well-known method for solving this problem involves regulating a set of inter-agent distances to values prescribed by the desired shape (de Queiroz, Cai, & Feemster, 2019; Krick, Broucke, & Francis, 2009). This method, which is commonly referred to as distance-based formation control (Oh, Park, & Ahn, 2015), has the main advantage of being implementable in a fully decentralized manner. However, this advantage comes with the limitation that the set of inter-agent distances may not uniquely define the formation position and orientation in space. Mathematically, this nonuniqueness is related to the existence of multiple equilibrium points in the multi-agent system distance dynamics. The question then is how do you steer the system away from the undesired equilibria and towards the equilibrium corresponding to the desired formation shape (up to translation and rotation).

The above question is partially answered by requiring the formation graph to be rigid, which imposes a minimum number of distances to be controlled. This reduces the unwanted equilibrium points to formations that are flipped or reflected versions of

the desired shape. Here, the agents' initial conditions determine whether the formation converges to one that is isomorphic to the desired formation or to a flipped/reflected formation. This implies that rigid distance-based controllers are locally stable.

In recent years, some methods have been introduced to address the limitation of distance-based formation controllers. The common feature of these methods is the use of an additional controlled variable (or constraint) that is capable of distinguishing formation ambiguities. In Mou, Cao, and Morse (2015), an approach called target-point control was introduced to rule out the undesirable equilibria in planar formations. The inter-agent distances and the order of agents were used to calculate the desired position of the agents, i.e., target position, in a local coordinate frame. However, the leader and the first follower cannot be collocated at time zero and the leader's motion needs to satisfy certain conditions. A similar method was used in Kang, Park, and Ahn (2017) with the name “desired order of neighbors”. In Ferreira-Vazquez, Hernandez-Martinez, Flores-Godoy, Fernandez-Anaya and Paniagua-Contro (2016), inter-agent distance and angular constraints were employed to enlarge the region of attraction to the desired planar formation by a proper choice of control gains. In Anderson, Sun, Sugie, Azuma, and Sakurama (2017), the signed area of a triangle was used as the second controlled variable, and convergence analyses were conducted for special cases of 3- and 4-agent planar formations. In Sugie, Anderson, Sun, and Dong (2018), the authors further explored the idea of Anderson et al. (2017) by applying the area constraints to only a subset of agents and thus extended the method to n agents. However, the result in Sugie et al. (2018) required the triangulated formation to be composed of equilateral triangles. Recently

[☆] The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Abdelhamid Tayebi under the direction of Editor Thomas Parisini.

* Corresponding author.

E-mail addresses: Tairan.Liu@uga.edu (T. Liu), mdeque1@lsu.edu (M. de Queiroz).

in Liu, de Queiroz, Zhang, and Khaledyan (2021), we extended the distance/signed area method to directed 2D formations of n agents and introduced the concept of strong congruency. In this result, the triangulations were not restricted to equilateral ones; however, certain triangulation and control gain conditions had to be met to prove the asymptotic stability of the desired formation. In Cao, Sun, Anderson, and Sugie (2019), a specific control gain value in the signed area term was found that causes the multi-agent system to have one stable equilibrium point corresponding to the desired formation and some discrete unstable equilibria. A switching control strategy based on the signed area or edge angle was proposed in Liu, Fernández-Kim and de Queiroz (2020) to remove the restrictions on the shape of the desired formation. In Liu and de Queiroz (2021), we designed a non-switching, distance/edge angle-based controller that ensures the almost-global asymptotic stability of the desired formation under certain conditions on the triangulations. Recently in Jing and Wang (2020), a formation controller based on angles and the “sign” of the triangulated framework was shown to guarantee almost-global convergence of the angle errors.

For the 3D formation problem, relatively few results exist. For example, Ferreira-Vazquez, Flores-Godoy, Hernandez-Martinez and Fernandez-Anaya (2016) extended the method in Ferreira-Vazquez, Hernandez-Martinez et al. (2016) to 3D by using distance and volume constraints. However, unless the control gains satisfy a persistency of excitation-type condition, the system under the control of Ferreira-Vazquez, Flores-Godoy et al. (2016) may still converge to an undesired formation shape. In Lan, Xu, and Wei (2018), volume constraints were applied to a 4-agent system to distinguish the two possible orientations of a tetrahedron under the assumption that three of the agents are at their desired distances.

In this paper, we address the problem of using additional feedback variables in the distance-based 3D formation controller. Specifically, we introduce a new method called the *orthogonal basis approach* which decomposes the feedback variables and control inputs into three orthogonal subspaces. By applying this decomposition to directed frameworks formed by tetrahedrons, we are able to guarantee the *global* asymptotic stability of the desired formation. Moreover, we can show the desired formation is locally exponentially stable which provides robustness to the system. These results are achieved with no limitations on the “tetrahedralizations” of the desired formation, control gains, or number of agents. Thus, this work greatly extends the applicability of the dual-feedback-variable approach for avoiding 3D formation ambiguities. To the best of our knowledge, it is the first result to show convergence to the desired 3D formation for all initial conditions, including collocated and collinear agents. A preliminary version of our orthogonal basis approach appeared in Liu, de Queiroz and Sahebsara (2020), where it was applied to planar formations and achieved almost-global asymptotic stability of the desired formation.

2. Background material

Some background material is reviewed in this section. Throughout the paper, we use the following vector notation: $x \in \mathbb{R}^n$ or $x = [x_1, \dots, x_n]$ denotes an $n \times 1$ (column) vector, and $x = [x_1, \dots, x_n]$ where $x_i \in \mathbb{R}^m$ is the stacked $mn \times 1$ vector.

2.1. Graph theory

An undirected graph G is represented by a pair $(\mathcal{V}, \mathcal{E}^u)$, where $\mathcal{V} = \{1, 2, \dots, N\}$ is the set of vertices and $\mathcal{E}^u = \{(i, j) | i, j \in \mathcal{V}, i \neq j\} \subset \mathcal{V} \times \mathcal{V}$ is the set of undirected edges. A directed graph G is a pair $(\mathcal{V}, \mathcal{E}^d)$ where the edge set $\mathcal{E}^d$ is directed in the sense

that if $(i, j) \in \mathcal{E}^d$ then i is the source vertex of the edge and j is the sink vertex. The set of neighbors of vertex $i \in \mathcal{V}$ is defined as $\mathcal{N}_i(\mathcal{E}^d) = \{j \in \mathcal{V} | (i, j) \in \mathcal{E}^d\}$. For $i \in \mathcal{V}$, the out-degree of i (denoted by $\text{out}(i)$) is the number of edges in $\mathcal{E}^d$ whose source is vertex i and sinks are in $\mathcal{V} - \{i\}$. If $p_i \in \mathbb{R}^3$ is the coordinate of the i th vertex of a 3D graph, then a framework F is defined as the pair (G, p) where $p = [p_1, \dots, p_N] \in \mathbb{R}^{3N}$.

Let the map $\mathcal{T} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be such that $\mathcal{T}(x) = \mathcal{R}x + d$ where $\mathcal{R} \in SO(3)$ and $d \in \mathbb{R}^3$. A framework $F = (G, p)$ is rigid in $\mathbb{R}^3$ if all of its continuous motions satisfy $p_i(t) = \mathcal{T}(p_i)$ for all $i = 1, \dots, N$ and $\forall t \geq 0$ (Asimow & Roth, 1979; Izemstiev, 2009). A 3D rigid framework is minimally rigid if and only if $|\mathcal{E}^u| = 3N - 6$ (Anderson, Yu, Fidan, & Hendrickx, 2008). The edge function of a minimally rigid framework $\gamma : \mathbb{R}^{3N} \rightarrow \mathbb{R}^{3N-3(3+1)/2}$ is defined as

$$\gamma(p) = [\dots, \|p_i - p_j\|, \dots], (i, j) \in \mathcal{E}^u \quad (1)$$

such that its l th component, $\|p_i - p_j\|$, relates to the l th edge of $\mathcal{E}^u$ connecting the i th and j th vertices. Frameworks (G, p) and $(G, \hat{p})$ are equivalent if $\gamma(p) = \gamma(\hat{p})$, and are congruent if $\|p_i - p_j\| = \|\hat{p}_i - \hat{p}_j\|$ for all distinct vertices i and j in $\mathcal{V}$ (Jackson, 2007). If rigid frameworks (G, p) and $(G, \hat{p})$ are equivalent but not congruent, they are flip- or flex-ambiguous (Anderson et al., 2008).

Frameworks based on directed graphs are required to be constraint consistent and persistent to maintain its shape (Yu, Hendrickx, Fidan, Anderson, & Blondel, 2007). A persistent graph is said to be minimally persistent if no single edge can be removed without losing persistence. A sufficient condition for a directed graph $(\mathcal{V}, \mathcal{E}^d)$ in $\mathbb{R}^3$ to be constraint consistent is $\text{out}(i) \leq 3$ for all $i \in \mathcal{V}$ (see Lemma 5 of Yu et al. (2007)). A necessary condition for a graph in $\mathbb{R}^3$ to be minimally persistent is $\text{out}(i) \leq 3$ for all $i \in \mathcal{V}$, while a sufficient condition is minimal rigidity (Yu et al., 2007). A minimally persistent graph can be constructed by the 3D Henneberg insertion of type I¹ (Grasgeger et al., 2018). This method starts with three vertices with three directed edges, and grows the graph by iteratively adding a vertex with three outgoing edges. Henceforth, we refer to a framework constructed in this manner as a 3D Henneberg framework.

2.2. Strong congruency

The concept of congruency defined above can distinguish between two frameworks that are flip- or flex-ambiguous, but does not capture a third type of ambiguity — a reflection of the whole framework. In Liu et al. (2021), we introduced the concept of *strong congruency* to handle this type of ambiguity in 2D. Specifically, Henneberg frameworks $F = (G, p)$ and $\hat{F} = (G, \hat{p})$ are said to be *strongly congruent* if they are congruent and not reflected versions of each other.

In Liu et al. (2021) (Lemma 2.2), it was shown that the signed area of a triangular framework in addition to a set of edge lengths can be used to ensure strong congruency in 2D. In order to extend this concept to 3D, we will employ the *signed volume of a tetrahedron*, $V : \mathbb{R}^{12} \rightarrow \mathbb{R}$ (Mallison, 1935):

$$\begin{aligned} V(p) &= \frac{1}{6} \det \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_1 & p_2 & p_3 & p_4 \end{bmatrix} \\ &= -\frac{1}{6} (p_1 - p_4)^T [(p_2 - p_4) \times (p_3 - p_4)] \end{aligned} \quad (2)$$

where $p = [p_1, p_2, p_3, p_4]$. If the order of vertices 1, 2, 3 is counterclockwise (resp., clockwise) from an observer located at vertex

¹ As shown in Grasgeger, Koutschan, and Tsigaridas (2018), the 3D Henneberg insertion of type I does not allow edge splitting operations (Eren, Whiteley, Anderson, Morse, & Belhumeur, 2005) in the graph construction procedure.

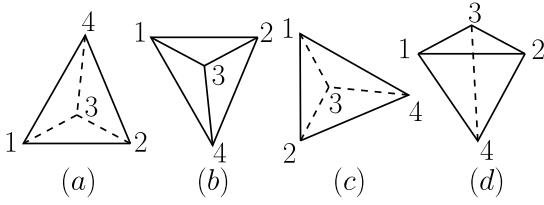


Fig. 1. All four 3D frameworks are congruent, but only the ones in (a), (b), and (c) are strongly congruent.

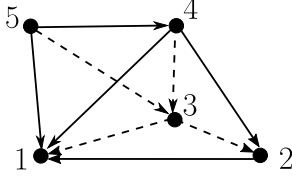


Fig. 2. Framework with two tetrahedrons.

4 facing the 1-2-3 plane, then (2) is positive (resp., negative). Moreover, this quantity is zero if any three vertices are collinear or the four vertices are coplanar. As an example of the use of the signed volume, all the frameworks in Fig. 1 are congruent, but only frameworks (a), (b), and (c) are strongly congruent. Hereafter, we denote the set of all 3-dimensional frameworks that are strongly congruent to framework F by $\text{SCgt}^3(F)$.

A 3D Henneberg framework can be divided into tetrahedral sub-frameworks. In such cases, the signed volume of a 3D Henneberg framework with N vertices and directed edge set $\mathcal{E}^d$, $\mathbf{V} : \mathbb{R}^{3N} \rightarrow \mathbb{R}^{N-3}$, is defined as

$$\mathbf{V}(p) = \left[\dots, \frac{1}{6} \det \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_i & p_j & p_k & p_l \end{bmatrix}, \dots \right], \quad \forall (l, i), (l, j), (l, k) \in \mathcal{E}^d - \{(2, 1), (3, 1), (3, 2)\} \quad (3)$$

where $p = [p_1, \dots, p_N]$ and its n th component is the signed volume of the n th tetrahedron constructed with vertices $i < j < k < l$. For example, the signed volume of the framework in Fig. 2 is given by

$$\mathbf{V}(p) = \begin{bmatrix} -\frac{1}{6}(p_1 - p_4)^T [(p_2 - p_4) \times (p_3 - p_4)] \\ -\frac{1}{6}(p_1 - p_5)^T [(p_3 - p_5) \times (p_4 - p_5)] \end{bmatrix} \quad (4)$$

where the first element is positive and the second one negative. Note that if $\hat{F}$ is a reflected version of F , then $\mathbf{V}(p) = -\mathbf{V}(\hat{p})$.

Lemma 1 (Liu & de Queiroz, 2020). 3D Henneberg frameworks $F = (G, p)$ and $\hat{F} = (G, \hat{p})$ are strongly congruent if and only if they are equivalent and $\mathbf{V}(p) = \mathbf{V}(\hat{p})$.

2.3. Stability results

Here, we recall some results concerning the stability of nonlinear systems.

Lemma 2 (Khalil, 2015). Suppose $f(x, u)$ is continuously differentiable and globally Lipschitz in $[x, u]$. If $\dot{x} = f(x, 0)$ has a globally exponentially stable (GES) equilibrium point at the origin, then the system $\dot{x} = f(x, u)$ is input-to-state stable (ISS).

Lemma 3 (Khalil, 2015). If the systems $\dot{\eta} = f_1(\eta, \xi)$ and $\dot{\xi} = f_2(\xi, u)$ are ISS, then the cascade connection

$$\dot{\eta} = f_1(\eta, \xi), \quad \dot{\xi} = f_2(\xi, u) \quad (5)$$

is ISS. Consequently, if $\dot{\eta} = f_1(\eta, \xi)$ is ISS and the origin of $\dot{\xi} = f_2(\xi, 0)$ is globally asymptotically stable (GAS), then the origin of the cascade connection

$$\dot{\eta} = f_1(\eta, \xi), \quad \dot{\xi} = f_2(\xi, 0) \quad (6)$$

is GAS.

3. Problem statement

Consider a system of mobile agents described by directed framework $F(t) = (G, p(t))$ where $G = (\mathcal{V}, \mathcal{E})$, $|\mathcal{V}| = N$, $|\mathcal{E}| = 3N - 6$, $N \geq 4$, $p = [p_1, \dots, p_N]$, and $p_i \in \mathbb{R}^3$ is the position of agent i . The directed edge $(j, i) \in \mathcal{E}$ means that agent j can measure its relative position to agent i , $p_{ji} := p_i - p_j$, but not the opposite. We assume agent j can sense all relative positions p_{ji} where $i \in \mathcal{N}_j(\mathcal{E})$. The agents are assumed to be governed by the dynamics

$$\dot{p}_i = u_i, \quad \forall i \in \mathcal{V} \quad (7)$$

where $u_i \in \mathbb{R}^3$ is the control input.

The desired formation is characterized by a set of desired distances d_{ji} , $(j, i) \in \mathcal{E}$ and a set of desired signed volumes $V_{ijkl}^* = V(p_1^*, p_j^*, p_k^*, p_l^*)$, $(l, i), (l, j), (l, k) \in \mathcal{E}$ (see (2)) where p_i^* is the desired position of agent i .² This gives the desired framework $F^* = (G, p^*)$ where $p^* = [p_1^*, \dots, p_N^*]$ and $\|p_j^* - p_i^*\| = d_{ji}$. We assume F^* satisfies the following conditions: i) F^* is non-degenerate in 3D space, i.e., all tetrahedrons have nonzero volume; ii) $\text{out}(i) = i - 1$ for $i = 1, 2, 3$ and $\text{out}(i) = 3$ for $i = 4, \dots, N$; and iii) if there is an edge between agents i and j , the direction must be $i \leftarrow j$ if $i < j$. Given these conditions, we say that agent 1 is the leader, agent 2 is the first follower, agent 3 is the secondary follower, and agents $i \geq 4$ are ordinary followers. Note that this nomenclature is the 3D extension of the leader–first-follower, minimally persistent framework discussed in Summers, Yu, Dasgupta, and Anderson (2011), which was called an acyclic minimally structural persistent framework in Lan et al. (2018).

Our control objective is to design u_i , $\forall i \in \mathcal{V}$ such that

$$F(t) \rightarrow \text{SCgt}^3(F^*) \text{ as } t \rightarrow \infty \quad (8)$$

for the largest set of initial conditions possible.

4. Orthogonal basis

Ambiguous frameworks in 3D can be discerned by employing the signed volume of the framework in the formation controller. However, this variable may introduce new undesired equilibria since the distance and volume constraints will interfere with each other at certain agent positions. In other words, these two variables do not always constitute an orthogonal space. To remedy this situation, we will introduce projection variables that are always orthogonal.

Consider the tetrahedron in Fig. 3. If $\|p_{ji}\| \neq 0$ and if there exists a vector n_{ijk} such that

$$n_{ijk}^T p_{ki} = 0 \quad \text{and} \quad n_{ijk}^T p_{kj} = 0, \quad (9)$$

then $p_{ji}^T n_{ijk} = 0$, $p_{ji}^T (n_{ijk} \times p_{ji}) = 0$, $n_{ijk}^T (n_{ijk} \times p_{ji}) = 0$, and we say $\{p_{ji}, n_{ijk} \times p_{ji}, n_{ijk}\}$ is the orthogonal basis of vertex i . With this in mind, the projection variables for vertex i are defined as

$$\zeta_i = p_{li}^T p_{ji} = \|p_{li}\|^2 - p_{li}^T p_{ji} \quad (10a)$$

$$\varphi_i = p_{li}^T (n_{ijk} \times p_{ji}) = p_{li}^T (n_{ijk} \times p_{ji}) \quad (10b)$$

² Note that p_i^* is not explicitly used by the control since we are not controlling the global position of the agents. This variable is only mentioned so we can formally define the desired formation.

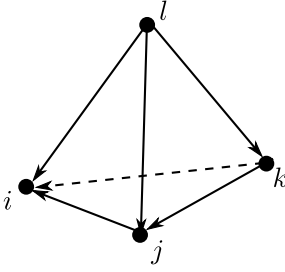


Fig. 3. Tetrahedron framework.

$$\vartheta_l = p_{li}^\top n_{ijk} = p_{lj}^\top n_{ijk} = p_{lk}^\top n_{ijk}, \quad (10c)$$

where

$$n_{ijk} = \begin{cases} p_{ki} \times p_{kj}, & \text{if } \{i, j, k\} \neq \{1, 2, 3\} \\ \frac{p_{31} \times p_{32}}{\|p_{31} \times p_{32}\|}, & \text{if } \{i, j, k\} = \{1, 2, 3\} \\ & \text{and } \|p_{31}(t) \times p_{32}(t)\| \neq 0 \\ n_{123}^+, & \text{if } \{i, j, k\} = \{1, 2, 3\} \\ & \text{and } \|p_{31}(0) \times p_{32}(0)\| = 0 \end{cases} \quad (11)$$

and n_{123}^+ is any unit vector satisfying $p_{31}^\top n_{123}^+ = 0$ and $p_{32}^\top n_{123}^+ = 0$. This unit vector can be computed using Algorithm 1. Notice that (10a) (resp., (10b); (10c)) is associated with the projection of p_{li} onto the direction of p_{ji} (resp., $n_{ijk} \times p_{ji}$; n_{ijk}). The reason for differentiating $\{i, j, k\} = \{1, 2, 3\}$ from the case $\{i, j, k\} \neq \{1, 2, 3\}$ in (11) is that our framework is composed of multiple tetrahedrons where vertices $\{1, 2, 3\}$ have different out-degree properties than the others (see Section 3).

Generally, the above projection variables are defined for agents $l \geq 4$ since a tetrahedron is composed of four vertices and only agents $l \geq 4$ have $\text{out}(l) = 3$. Therefore, special definitions for the projection variables are required for agents 2 and 3. For agent 2, we define

$$\zeta_2 = p_{21}^\top n_2, \quad (12)$$

where

$$n_2 = \begin{cases} \frac{p_{21}}{\|p_{21}\|}, & \text{if } \|p_{21}(t)\| \neq 0 \\ n_2^+, & \text{if } \|p_{21}(0)\| = 0 \end{cases} \quad (13)$$

and n_2^+ is any unit vector. Variables φ_2 and ϑ_2 are undefined since $\text{out}(2) = 1$. For agent 3, we let $l = k = 3$, $i = 1$, and $j = 2$ in (10a) and (10b) to obtain

$$\zeta_3 = p_{31}^\top p_{21} = \|p_{31}\|^2 - p_{31}^\top p_{32} \quad (14)$$

and

$$\varphi_3 = p_{31}^\top (n_{123} \times p_{21}). \quad (15)$$

Here, variable ϑ_3 is undefined since $\text{out}(3) = 2$.

Remark 1. One can show that projection variable (10c) is related to the signed volume of a tetrahedron and the area of a triangle. Consider (10c) when $\{i, j, k\} \neq \{1, 2, 3\}$. It follows from (11) that

$$\begin{aligned} \vartheta_l &= p_{li}^\top n_{ijk} = p_{li}^\top (p_{ki} \times p_{kj}) \\ &= p_{li}^\top (p_{ki} \times p_{kj}) - p_{lk}^\top (p_{ki} \times p_{kj}) = p_{lk}^\top (p_{ki} \times p_{kj}) \end{aligned}$$

Algorithm 1 Selecting n_{123} when $\|p_{31}(0) \times p_{32}(0)\| = 0$. (The value $1e-3$ below can be replaced with any sufficiently small number.)

```

1: Input:  $p_{31} = [x_1, y_1, z_1]$ ,  $p_{32} = [x_2, y_2, z_2]$ 
2: Output:  $n_{123}$ 
3:  $\text{eps} \leftarrow 1e-3$ 
4: if  $\|p_{31}\| > \text{eps}$  then
5:   if  $|z_1| > \text{eps}$  then
6:      $n_z \leftarrow -(x_1 + y_1) / z_1$ 
7:      $n_t \leftarrow [1, 1, n_z]$ 
8:   else if  $|y_1| > \text{eps}$  then
9:      $n_y \leftarrow -(x_1 + z_1) / y_1$ 
10:     $n_t \leftarrow [1, n_y, 1]$ 
11:   else if  $|x_1| > \text{eps}$  then
12:      $n_x \leftarrow -(y_1 + z_1) / x_1$ 
13:      $n_t \leftarrow [n_x, 1, 1]$ 
14:   end if
15: else if  $\|p_{32}\| > \text{eps}$  then
16:   if  $|z_2| > \text{eps}$  then
17:      $n_z \leftarrow -(x_2 + y_2) / z_2$ 
18:      $n_t \leftarrow [1, 1, n_z]$ 
19:   else if  $|y_2| > \text{eps}$  then
20:      $n_y \leftarrow -(x_2 + z_2) / y_2$ 
21:      $n_t \leftarrow [1, n_y, 1]$ 
22:   else if  $|x_2| > \text{eps}$  then
23:      $n_x \leftarrow -(y_2 + z_2) / x_2$ 
24:      $n_t \leftarrow [n_x, 1, 1]$ 
25:   end if
26: else
27:    $n_t \leftarrow [0, 0, 1]$ 
28: end if
29:  $n_{123} \leftarrow n_t / \|n_t\|$ 

```

$$\begin{aligned} &= p_{lk}^\top [(p_{li} - p_{lk}) \times (p_{lj} - p_{lk})] \\ &= p_{lk}^\top [p_{li} \times p_{lj} - p_{li} \times p_{lk} - p_{lk} \times p_{lj} + p_{lk} \times p_{lk}] \\ &= p_{lk}^\top (p_{li} \times p_{lj}) = p_{li}^\top (p_{lj} \times p_{lk}) = -6V_{ijkl} \end{aligned} \quad (16)$$

where $V_{ijkl} := V(p_i, p_j, p_k, p_l)$ from (2). When $\{i, j, k\} = \{1, 2, 3\}$ and $\|p_{31} \times p_{32}\| \neq 0$, we have

$$\vartheta_l = p_{l1}^\top n_{123} = \frac{p_{l1}^\top (p_{31} \times p_{32})}{\|p_{31} \times p_{32}\|} = -\frac{3V_{123l}}{\check{S}_{123}} \quad (17)$$

where $\check{S}_{ijk}$ is the regular (unsigned) area of $\triangle ijk$ (Heron's formula Zwillinger, 2002):

$$\begin{aligned} \check{S}_{ijk} &= \frac{1}{2} \|p_{ki} \times p_{kj}\| \\ &= \frac{1}{4} \left(2 \|p_{ji}\|^2 \|p_{ki}\|^2 + 2 \|p_{ji}\|^2 \|p_{kj}\|^2 \right. \\ &\quad \left. + 2 \|p_{ki}\|^2 \|p_{kj}\|^2 - \|p_{ji}\|^4 - \|p_{ki}\|^4 - \|p_{kj}\|^4 \right)^{1/2}. \end{aligned} \quad (18)$$

The projection variables of a tetrahedralized Henneberg framework (G, p) with $|\mathcal{V}| = N$ is defined as

$$\Lambda(p) = [\Lambda_2, \Lambda_3, \Lambda_4, \dots, \Lambda_N] \quad (19)$$

where

$$\begin{aligned} \Lambda_2 &= \zeta_2 \\ \Lambda_3 &= [\zeta_3, \varphi_3] \\ \Lambda_4 &= [\zeta_4, \varphi_4, \vartheta_4] \\ &\vdots \\ \Lambda_N &= [\zeta_N, \varphi_N, \vartheta_N]. \end{aligned} \quad (20)$$

Lemma 4 (Liu & de Queiroz, 2020). 3D Henneberg frameworks $F = (G, p)$ and $\hat{F} = (G, \hat{p})$ are strongly congruent if and only if $\Lambda(p) = \Lambda(\hat{p})$.

Note that by Lemma 4, the control objective in (8) is equivalent to

$$\Lambda(p(t)) \rightarrow \Lambda(p^*) \text{ as } t \rightarrow \infty. \quad (21)$$

5. Formation controller

5.1. Error variables

The control objective will be quantified by the following three projection error variables

$$\tilde{\zeta}_l = \zeta_l - \zeta_l^* \quad (22a)$$

$$\tilde{\varphi}_l = \varphi_l - \varphi_l^* \quad (22b)$$

$$\tilde{\vartheta}_l = \vartheta_l - \vartheta_l^* \quad (22c)$$

where the asterisk denotes the desired value for the projection. Since F^* is typically specified in terms of the desired inter-agent distances, the desired projections can be calculated in terms of d_{ji} , $(j, i) \in \mathcal{E}$ as shown in Appendix. The stacked vector of all the projection errors is represented by $\tilde{A} = [\tilde{A}_2, \tilde{A}_3, \dots, \tilde{A}_N]$ where $\tilde{A}_2 = \tilde{\zeta}_2$, $\tilde{A}_3 = [\tilde{\zeta}_3, \tilde{\varphi}_3]$, and $\tilde{A}_i = [\tilde{\zeta}_i, \tilde{\varphi}_i, \tilde{\vartheta}_i]$ for $i = 4, \dots, N$.

Lemma 5 (Liu & de Queiroz, 2020). For agent l , the planes corresponding to $\tilde{\zeta}_l = 0$, $\tilde{\varphi}_l = 0$, and $\tilde{\vartheta}_l = 0$ are mutually orthogonal if $\|p_{ji}\| = d_{ji}$, $\|p_{ki}\| = d_{ki}$, and $\|p_{kj}\| = d_{kj}$, where $i < j < k < l$ and $(l, i), (l, j), (l, k) \in \mathcal{E}$.

5.2. Control law

We propose the following formation control

$$u_1 = 0 \quad (23a)$$

$$u_2 = \mu_2 \tilde{\zeta}_2 n_2 \quad (23b)$$

$$u_3 = \mu_3 \tilde{\zeta}_3 (p_{31} - p_{32}) + v_3 \tilde{\varphi}_3 n_{123} \times (p_{31} - p_{32}) \quad (23c)$$

$$u_l = \mu_l \tilde{\zeta}_l (p_{li} - p_{lj}) + v_l \tilde{\varphi}_l n_{ijk} \times (p_{li} - p_{lj}) + \lambda_l \tilde{\vartheta}_l n_{ijk}, \quad l = 4, \dots, N \quad (23d)$$

where $i < j < k < l$, $(l, i), (l, j), (l, k) \in \mathcal{E}$, and μ_l, v_l, λ_l are positive control gains.

Note that since $p_{ki} = p_i - p_k$ and $p_{kj} = p_j - p_k$ (see Fig. 3), the term n_{ijk} defined in (11) can be expressed as a function of p_{li} , p_{lj} , and p_{lk} for $(l, i), (l, j), (l, k) \in \mathcal{E}$. Therefore, the above control is decentralized in the sense that it requires each agent to know its relative position to neighboring agents only. This means that control (23) can be implemented in each agent's local coordinate frame.

Remark 2. Since a 3D minimally persistent graph has $3N - 6$ edges, a 3D formation requires $3N - 6$ constraints to maintain its shape. Previous 3D formation control work that utilized distance and volume variables (Ferreira-Vazquez, Flores-Godoy et al., 2016) are over-constrained since they require $3N - 6$ distance constraints and $N - 3$ volume constraints. Although these extra constraints rule out formation ambiguities, they introduce new undesired equilibria. On the other hand, the orthogonal basis method proposed here has $N - 1$ ζ -type projection constraints, $N - 2$ φ -type projection constraints, and $N - 3$ ϑ -type projection constraints, totaling exactly $3N - 6$ constraints.

5.3. Preliminary analysis

In preparation for our main results, we present next some preliminary results.

Lemma 6. With control (23), $\dot{n}_2 = 0$ and $\dot{n}_{123} = 0$.

Proof. If $\|p_{21}\| \neq 0$, the time derivative of (13) is given by

$$\begin{aligned} \dot{n}_2 &= \frac{1}{\|p_{21}\|} (\dot{p}_1 - \dot{p}_2) - \frac{p_{21}}{\|p_{21}\|^2} \frac{d}{dt} \|p_{21}\| \\ &= \frac{1}{\|p_{21}\|} (\dot{p}_1 - \dot{p}_2) - \frac{p_{21}}{\|p_{21}\|^2} \frac{p_{21}^\top (\dot{p}_1 - \dot{p}_2)}{\|p_{21}\|} \\ &= -\frac{1}{\|p_{21}\|} u_2 + \frac{p_{21}}{\|p_{21}\|^3} p_{21}^\top u_2. \end{aligned} \quad (24)$$

After substituting (23) into (24), we obtain

$$\begin{aligned} \dot{n}_2 &= -\frac{1}{\|p_{21}\|} \left(\mu_2 \tilde{\zeta}_2 \frac{p_{21}}{\|p_{21}\|} \right) + \frac{p_{21}}{\|p_{21}\|^3} p_{21}^\top \left(\mu_2 \tilde{\zeta}_2 \frac{p_{21}}{\|p_{21}\|} \right) \\ &= -\mu_2 \tilde{\zeta}_2 \frac{p_{21}}{\|p_{21}\|^2} + \mu_2 \tilde{\zeta}_2 \frac{p_{21}}{\|p_{21}\|^2} \\ &= 0. \end{aligned} \quad (25)$$

When $\|p_{21}\| = 0$, it is obvious from (13) that $\dot{n}_2 = 0$.

If $\|p_{31} \times p_{32}\| \neq 0$, the derivative of (11) is given by

$$\begin{aligned} \dot{n}_{123} &= \frac{1}{\|p_{31} \times p_{32}\|} \frac{d}{dt} (p_{31} \times p_{32}) \\ &\quad - (p_{31} \times p_{32}) \frac{(p_{31} \times p_{32})^\top}{\|p_{31} \times p_{32}\|^3} \frac{d}{dt} (p_{31} \times p_{32}) \end{aligned} \quad (26)$$

where

$$\frac{d}{dt} (p_{31} \times p_{32}) = (u_1 - u_3) \times p_{32} + p_{31} \times (u_2 - u_3). \quad (27)$$

Substituting (23) into (27) yields

$$\begin{aligned} &\frac{d}{dt} (p_{31} \times p_{32}) \\ &= - (p_{31} - p_{32}) \times (\mu_3 \tilde{\zeta}_3 p_{21} + v_3 \tilde{\varphi}_3 n_{123} \times p_{21}) \\ &\quad + p_{31} \times (\mu_2 \tilde{\zeta}_2 n_2) \\ &= -v_3 \tilde{\varphi}_3 \|p_{21}\|^2 n_{123} + \mu_2 \tilde{\zeta}_2 p_{31} \times n_2. \end{aligned} \quad (28)$$

Since $\|p_{31} \times p_{32}\| \neq 0$, we have

$$\|p_{31} \times p_{32}\| = \|p_{31} \times (p_{31} - p_{21})\| = \|p_{31} \times p_{21}\| \neq 0,$$

which implies $\|p_{21}\| \neq 0$. Then, we can substitute the first case of (13) and the second case of (11) into (28) to obtain

$$\begin{aligned} &\frac{d}{dt} (p_{31} \times p_{32}) \\ &= -v_3 \tilde{\varphi}_3 \|p_{21}\|^2 \frac{p_{31} \times p_{32}}{\|p_{31} \times p_{32}\|} + \mu_2 \tilde{\zeta}_2 p_{31} \times \frac{p_{21}}{\|p_{21}\|} \\ &= -v_3 \tilde{\varphi}_3 \|p_{21}\|^2 \frac{p_{31} \times p_{32}}{\|p_{31} \times p_{32}\|} - \mu_2 \tilde{\zeta}_2 \frac{p_{31} \times p_{32}}{\|p_{21}\|}. \end{aligned} \quad (29)$$

Now, substituting (29) into (26) gives

$$\begin{aligned} \dot{n}_{123} &= \frac{1}{\|p_{31} \times p_{32}\|} \\ &\quad \cdot \left(-v_3 \tilde{\varphi}_3 \|p_{21}\|^2 \frac{p_{31} \times p_{32}}{\|p_{31} \times p_{32}\|} - \mu_2 \tilde{\zeta}_2 \frac{p_{31} \times p_{32}}{\|p_{21}\|} \right) \\ &\quad - (p_{31} \times p_{32}) \cdot \frac{(p_{31} \times p_{32})^\top}{\|p_{31} \times p_{32}\|^3} \left(-v_3 \tilde{\varphi}_3 \|p_{21}\|^2 \right. \\ &\quad \cdot \frac{p_{31} \times p_{32}}{\|p_{31} \times p_{32}\|} - \mu_2 \tilde{\zeta}_2 \frac{p_{31} \times p_{32}}{\|p_{21}\|} \left. \right) \\ &= 0. \end{aligned} \quad (30)$$

For $p_{31} \times p_{32} = 0$, it is clear from (11) that $\dot{n}_{123} = 0$. $\square$

Remark 3. The purpose of n_2^+ in (13) is to force agent 2 to leave the collocated initial position with agent 1. Since $\dot{n}_2 = 0$, the two

cases of (13) have the same value. For example, assume the two agents are collocated at time zero and let $n_2^+ = [1, 0, 0]$. Control (23b) will cause agent 2 to move away from the collocated position along vector $[1, 0, 0]$. Since the agents are no longer collocated, then $\frac{p_{21}}{\|p_{21}\|} = [1, 0, 0]$. The vector n_{123}^+ in (11) serves a

similar purpose, viz., to force agent 3 to leave the collinear initial position with agents 1 and 2. Note that the second and third cases of (11) have the same value due to $\dot{n}_{123} = 0$. If the three agents are collinear at time zero on the x - y plane and $n_{123}^+ = [0, 0, 1]$ for example, then the second term in (23c) will cause agent 3 to move on the x - y plane away from the collinear position.

Lemma 7. For the two-agent system $\{1, 2\}$, (23a) and (23b) render $\dot{\tilde{\zeta}}_2 = 0$ GES.

Proof. The time derivative of $\tilde{\zeta}_2$ is given by

$$\dot{\tilde{\zeta}}_2 = (\dot{p}_1 - \dot{p}_2)^T n_2 + p_{21}^T \dot{n}_2. \quad (31)$$

Applying Lemma 6 and substituting (23a) and (23b) into (31), we obtain

$$\dot{\tilde{\zeta}}_2 = -n_2^T u_2 = -\mu_2 \tilde{\zeta}_2 \|n_2\|^2 = -\mu_2 \tilde{\zeta}_2 \quad (32)$$

which indicates that $\tilde{\zeta}_2 = 0$ is GES. $\square$

Now, consider the Lyapunov function candidates

$$W_3 = \frac{1}{2} \tilde{\zeta}_3^2 + \frac{1}{2} \tilde{\varphi}_3^2 \quad (33a)$$

$$W_l = \frac{1}{2} \tilde{\zeta}_l^2 + \frac{1}{2} \tilde{\varphi}_l^2 + \frac{1}{2} \tilde{\vartheta}_l^2, \quad l = 4, \dots, N \quad (33b)$$

where $i < j < k < l, (l, i), (l, j), (l, k) \in \mathcal{E}$.

Lemma 8. If $\|p_{21}\| = d_{21}$ and $u_2 = 0$, then (23a) and (23c) render $\tilde{\Lambda}_3 = 0$ GES for the three-agent system $\{1, 2, 3\}$.

Proof. The dynamics of $\tilde{\zeta}_3$ is given by

$$\dot{\tilde{\zeta}}_3 = \dot{p}_{31}^T p_{21} + p_{31}^T \dot{p}_{21} = (u_1 - u_3)^T p_{21} + p_{31}^T (u_1 - u_2) \quad (34)$$

where (22a), (10a), and (7) were used. Similarly, the dynamics of $\tilde{\varphi}_3$ can be computed as

$$\dot{\tilde{\varphi}}_3 = (u_1 - u_3)^T (n_{123} \times p_{21}) + p_{31}^T (\dot{n}_{123} \times p_{21}) + p_{31}^T (n_{123} \times (u_1 - u_2)). \quad (35)$$

Therefore, the time derivative of (33a) is given by

$$\begin{aligned} \dot{W}_3 &= \dot{\tilde{\zeta}}_3 \tilde{\zeta}_3 + \dot{\tilde{\varphi}}_3 \tilde{\varphi}_3 \\ &= \tilde{\zeta}_3 [p_{21}^T (u_1 - u_3) + p_{31}^T (u_1 - u_2)] \\ &\quad + \tilde{\varphi}_3 [(n_{123} \times p_{21})^T (u_1 - u_3) \\ &\quad + p_{31}^T (\dot{n}_{123} \times p_{21}) + p_{31}^T (n_{123} \times (u_1 - u_2))]. \end{aligned} \quad (36)$$

Recall from Lemma 6 that $\dot{n}_{123} = 0$. Therefore, substituting $u_2 = 0$, $\|p_{21}\| = d_{21}$, (23a), and (23c) into (36) gives

$$\begin{aligned} \dot{W}_3 &= -[\tilde{\zeta}_3 p_{21}^T + \tilde{\varphi}_3 (n_{123} \times p_{21})^T] u_3 \\ &= -\mu_3 \tilde{\zeta}_3^2 d_{21}^2 - \nu_3 \tilde{\varphi}_3^2 \|n_{123} \times p_{21}\|^2 \end{aligned} \quad (37)$$

where we used the fact that $p_{31} - p_{32} = p_{21}$. Since $\|n_{123}\| = 1$, then $\|n_{123} \times p_{21}\| = d_{21}$ and (37) becomes

$$\dot{W}_3 = -d_{21}^2 (\mu_3 \tilde{\zeta}_3^2 + \nu_3 \tilde{\varphi}_3^2), \quad (38)$$

which means $\tilde{\Lambda}_3 = 0$ is GES. $\square$

Lemma 9. If $\|p_{ji}\| = d_{ji}$, $\|p_{ki}\| = d_{ki}$, $\|p_{kj}\| = d_{kj}$, and $u_i = u_j = u_k = 0$, then (23d) ensures that $\tilde{\Lambda}_l = [\tilde{\zeta}_l, \tilde{\varphi}_l, \tilde{\vartheta}_l] = 0$ is GES for the tetrahedron formed by agents $\{i, j, k, l\}$.

Proof. The time derivative of (33b) along the dynamics of (22a), (22b), and (22c) is given by

$$\begin{aligned} \dot{W}_l &= \tilde{\zeta}_l [p_{ji}^T (u_i - u_l) + p_{li}^T (u_i - u_j)] \\ &\quad + \tilde{\varphi}_l [(u_i - u_l)^T (n_{ijk} \times p_{ji}) + p_{li}^T (\dot{n}_{ijk} \times p_{ji}) \\ &\quad + p_{li}^T (n_{ijk} \times (u_i - u_j))] \\ &\quad + \tilde{\vartheta}_l [\dot{n}_{ijk}^T (u_i - u_l) + p_{li}^T \dot{n}_{ijk}] \end{aligned} \quad (39)$$

where

$$\dot{n}_{ijk} = (u_i - u_k) \times p_{kj} + p_{ki} \times (u_j - u_k) \text{ if } \{i, j, k\} \neq \{1, 2, 3\},$$

and $\dot{n}_{123} = 0$ from Lemma 6.

After substituting $\|p_{ji}\| = d_{ji}$, $u_i = u_j = u_k = 0$, and (23d) into (39), we obtain

$$\begin{aligned} \dot{W}_l &= -[\tilde{\zeta}_l p_{ji}^T + \tilde{\varphi}_l (n_{ijk} \times p_{ji})^T + \tilde{\vartheta}_l n_{ijk}^T] u_l \\ &= -\mu_l d_{ji}^2 \tilde{\zeta}_l^2 - \nu_l \tilde{\varphi}_l^2 \|n_{ijk} \times p_{ji}\|^2 - \lambda_l \tilde{\vartheta}_l^2 \|n_{ijk}\|^2 \end{aligned} \quad (40)$$

where we used the fact that $p_{ji}^T n_{ijk} = 0$. Given that $\|p_{ki}\| = d_{ki}$ and $\|p_{kj}\| = d_{kj}$, we know from (11) and (18) that $\|n_{ijk}\|$ is constant. Therefore,

$$\dot{W}_l = -\mu_l d_{ji}^2 \tilde{\zeta}_l^2 - \nu_l d_{ji}^2 c \tilde{\varphi}_l^2 - \lambda_l c \tilde{\vartheta}_l^2 \quad (41)$$

where c is some positive constant. It then follows from (41) and (33b) that $[\tilde{\zeta}_l, \tilde{\varphi}_l, \tilde{\vartheta}_l] = 0$ is GES. $\square$

5.4. Main results

The following two theorems give our main results.

Theorem 1. Control (23) ensures $\tilde{\Lambda} = 0$ is GAS and $F(t) \rightarrow SCgt^3(F^*)$ as $t \rightarrow \infty$.

Proof. We know from Lemma 7 that the subsystem composed of agents 1 and 2 is GES at $\tilde{\zeta}_2 = 0$. If a third agent is added to this subsystem, we get the cascade system

$$\dot{\tilde{\Lambda}}_3 = f_3(\tilde{\Lambda}_3, \tilde{\zeta}_2) \quad (42a)$$

$$\dot{\tilde{\zeta}}_2 = g_2(\tilde{\zeta}_2) \quad (42b)$$

where (42b) is in fact (32). If $\tilde{\zeta}_2 = 0$, then $u_2 = 0$ from (23b) and $\|p_{21}\| = d_{21}$ from (20) and (22a). Therefore, (42a) with $\tilde{\zeta}_2 = 0$ is GES at the origin by Lemma 8. It then follows from Lemma 2 that (42a) is ISS with respect to input $\tilde{\zeta}_2$. Finally, we can use Lemma 3 to show that the origin of (42), i.e., $[\tilde{\zeta}_2, \tilde{\Lambda}_3] = 0$, is GAS.

As we grow the graph step-by-step in the analysis by adding a vertex l with three outgoing edges to any distinct vertices i, j and k of the previous graph, we obtain the following cascade system at each step:

$$\dot{\tilde{\Lambda}}_l = f_l(\tilde{\Lambda}_l, z_{l-1}) \quad (43a)$$

$$\dot{z}_{l-1} = g_{l-1}(z_{l-1}) \quad (43b)$$

where $z_{l-1} = [\tilde{\Lambda}_2, \dots, \tilde{\Lambda}_{l-1}]$ and $i < j < k < l$. Note that the GAS of $z_{l-1} = 0$ for (43b) was established in the previous step. Therefore, we only need to check if (43a) is ISS with respect to input z_{l-1} . If $z_{l-1} = 0$, then $u_i = u_j = u_k = 0$ from (23d). Now, consider subframeworks $F_{l-1} = (G_{l-1}, p)$ and $F_{l-1}^* = (G_{l-1}, p^*)$ where G_{l-1} is the subgraph of G that contains vertices $\{1, \dots, l-1\}$ and the corresponding edges connecting these vertices in the original graph. The condition $z_{l-1} = 0$ is equivalent to $[\Lambda_2^*, \dots, \Lambda_{l-1}^*] = [\Lambda_2^*, \dots, \Lambda_{l-1}^*]$ where $\Lambda_i^* = \Lambda_i(\tilde{p}^*)$, and $\tilde{p}^* = [p_1^*, \dots, p_{l-1}^*]$. This indicates that F_{l-1} and F_{l-1}^* are strongly congruent from Lemma 4. Thus, we know $\|p_{ji}\| = d_{ji}$, $\|p_{ki}\| = d_{ki}$, and $\|p_{kj}\| = d_{kj}$ from Lemma 1. We can now use Lemma 9 to show that (43a) with $z_{l-1} = 0$ is GES at the origin. As a result, (43a) is

ISS by Lemma 2. Finally, we can invoke Lemma 3 to conclude that $[z_{l-1}, \tilde{\Lambda}_l] = 0$ in (43) is GAS.

Repeating this process until $l = N$ leads to the conclusion that $\tilde{\Lambda} = 0$ is GAS, which implies $\Lambda(p(t)) \rightarrow \Lambda(p^*)$ as $t \rightarrow \infty$. Thus, we know from Lemma 4 that $F(t) \rightarrow \text{SCgt}^3(F^*)$ as $t \rightarrow \infty$. $\square$

Next, we show that the proposed control yields *local exponential* convergence to the equilibrium point. This property is important in practice since exponential stability is known to provide some level of robustness to the system (Khalil, 2015).

Theorem 2. *In the neighborhood of $\tilde{\Lambda} = 0$, the equilibrium point is locally exponentially stable (LES).*

Proof. The error dynamics for $\tilde{\Lambda}$ can be expressed as

$$\dot{\tilde{\Lambda}} = -A(\tilde{\Lambda})\tilde{\Lambda} \quad (44)$$

where

$$A(\tilde{\Lambda}) = \begin{bmatrix} D_1 & 0 & 0 & 0 & \cdots & \cdots & 0 \\ \star & D_2 & 0 & 0 & \cdots & \cdots & 0 \\ \star & 0 & D_3 & 0 & \cdots & \cdots & 0 \\ \star & \star & \star & D_4 & 0 & 0 & 0 \\ \star & \star & \star & 0 & D_5 & 0 & 0 \\ \star & \star & \star & 0 & 0 & D_6 & 0 \\ \vdots & & & & \ddots & & \vdots \\ \star & \cdots & \cdots & \star & \star & 0 & 0 \\ \star & \cdots & \cdots & \star & 0 & \star & 0 \\ \star & \cdots & \cdots & \star & 0 & 0 & D_{3N-6} \end{bmatrix} \quad (45)$$

is a lower triangular $[3(N-3)+3] \times [3(N-3)+3]$ matrix whose diagonal elements are $D_1 = \mu_2$, $D_2 = \mu_3 \|p_{21}\|^2$, $D_3 = \nu_3 \|n_{123} \times p_{21}\|^2$, $D_4 = \mu_4 \|p_{21}\|^2$, $D_5 = \nu_4 \|n_{123} \times p_{21}\|^2$, $D_6 = \lambda_4 \|n_{123}\|^2$, ..., $D_{3N-8} = \mu_N \|p_{N_1 N_2}\|^2$, $D_{3N-7} = \nu_N \|n_{N_1 N_2 N_3} \times p_{N_1 N_2}\|^2$, and $D_{3N-6} = \lambda_N \|n_{N_1 N_2 N_3}\|^2$ where N_1, N_2 , and N_3 ($N_1 < N_2 < N_3$) are the out-neighbors of agent N .

Linearizing (44) at the equilibrium $\tilde{\Lambda} = 0$ yields

$$\dot{\tilde{\Lambda}} \approx -A(0)\tilde{\Lambda} \quad (46)$$

where $A(0)$ is a constant matrix whose eigenvalues (diagonal elements) are positive and can be made arbitrarily large by adjusting the control gains. Therefore, $\tilde{\Lambda} = 0$ is LES. $\square$

Remark 4. It is worth noting that the 2D formation problem can be viewed as a degenerate case of the 3D problem. Specifically, we can consider the coordinates of agent i as $p_i = [x_i, y_i, 0]$ and express the control law with the third component equal to zero, i.e., $u_i = [u_{ix}, u_{iy}, 0]$. The results in Theorems 1 and 2 are also valid in the 2D case. This analysis is omitted here since it is based on similar arguments as above.

6. Numerical validation

Control law (23) was evaluated via simulations conducted in MATLAB. Eight agents were collocated at position $[0, 0, 0]$ at $t = 0$, which is a situation where the distance-only formation controller and previous augmented distance-based formation controllers do not ensure convergence to the desired formation. The control objective was for the agents to form the framework described by the unit cube in Fig. 4. The 18 directed edges shown in Fig. 4 were used to ensure the framework was minimally persistent. Thus, the edge set was $\mathcal{E} = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), (5, 2), (5, 3), (5, 4), (6, 3), (6, 4), (6, 5), (7, 4), (7, 5), (7, 6), (8, 2), (8, 4), (8, 7)\}$. The desired distances of the agents were $d_{21} = d_{31} = d_{41} = d_{52} =$

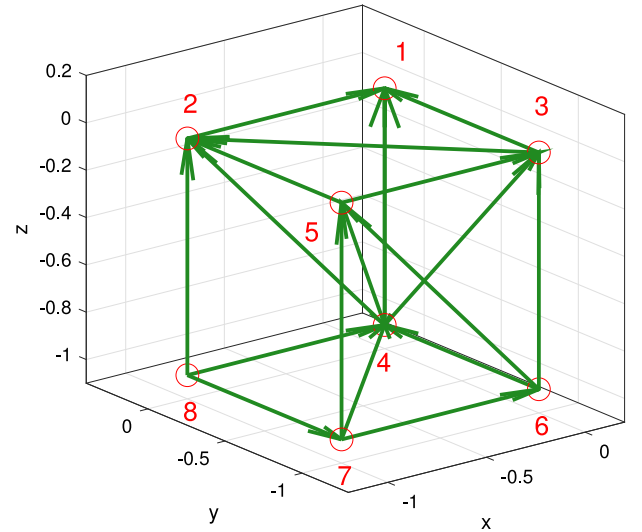


Fig. 4. Desired formation shape.

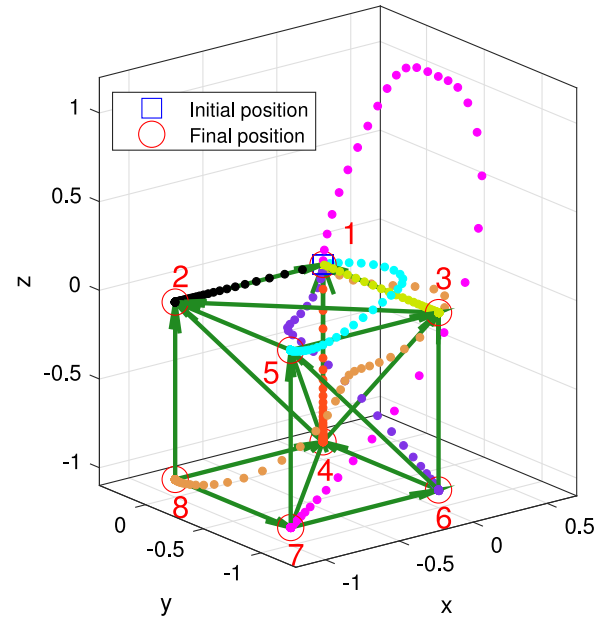


Fig. 5. Agent trajectories $p_i(t)$, $i = 1, \dots, 8$ for $t = [0, 3]$. At $t = 0$, agents are collocated at position $[0, 0, 0]$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$d_{53} = d_{63} = d_{64} = d_{75} = d_{76} = d_{82} = d_{84} = d_{87} = 1$, $d_{32} = d_{42} = d_{43} = d_{65} = d_{74} = \sqrt{2}$, and $d_{54} = \sqrt{3}$. The five desired signed volumes were $V_{1234}^* = V_{2345}^* = V_{3456}^* = V_{4567}^* = -1/6$ and $V_{2478}^* = 1/6$. The control gains in (23) were all set to 3. The trajectory of each agent from the initial position to the moment where the desired formation is acquired is shown in Fig. 5. We use the blue square to represent the agents' collocated initial positions, red circles to represent their final positions, and different color of dots to represent their trajectories. Notice that agent 1 remains stationary at position $[0, 0, 0]$ due to (23a) and serves as an "anchor" for the formation. The projection error variables defined in (22a), (22b), and (22c) are shown in Fig. 6. The results show the agents successfully converging to the desired formation after approximately 2 units of time despite their collocated initial positions.

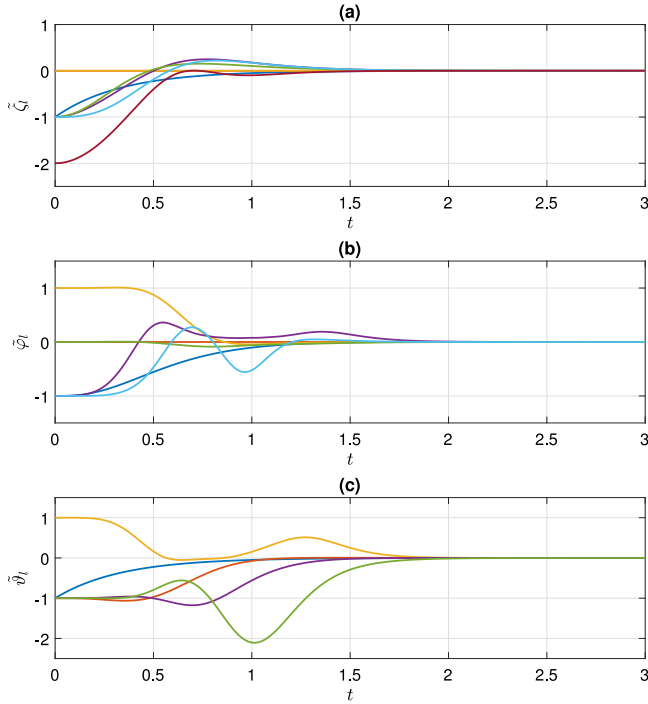


Fig. 6. Projection error variables $\tilde{\zeta}_l(t)$, $\tilde{\varphi}_l(t)$, and $\tilde{\vartheta}_l(t)$.

7. Conclusion

This paper introduced a new set of controlled variables for avoiding undesirable equilibria in the 3D distance-based formation control approach. The proposed variables, which are related to the inter-agent distances, signed volume of the framework, and areas of the triangular faces, form an orthogonal basis that decomposes the control inputs into orthogonal subspaces. The resulting formation controller guarantees the global asymptotic stability and the local exponential stability of the desired formation shape with any initial conditions. This result is valid for any tetrahedralized-like framework with no conditions on the formation shape or control gains.

The proposed approach can also handle the 2D formation problem, unifying the two problems. Specifically, by setting the z-coordinate of each agent position to zero, we arrive at the orthogonal basis controller for 2D formations that appeared in our preliminary result in Liu, de Queiroz et al. (2020).

Appendix. Desired projection variables

In the following, we show how the desired values for the 3D projection variables can be computed in terms of the desired inter-agent distances.

First, we have from (10a) that

$$\begin{aligned} \zeta_l^* &= (p_{li}^*)^\top p_{ji}^* \\ &= \|p_{li}^*\| \|p_{ji}^*\| \frac{\|p_{li}^*\|^2 + \|p_{ji}^*\|^2 - \|p_{ij}^*\|^2}{2 \|p_{li}^*\| \|p_{ji}^*\|} \\ &= \frac{d_{li}^2 + d_{ji}^2 - d_{ij}^2}{2}. \end{aligned} \quad (47)$$

Next, from (10b):

$$\begin{aligned} \varphi_l^* &= (p_{li}^*)^\top (n_{ijk}^* \times p_{ji}^*) = (n_{ijk}^*)^\top (p_{ji}^* \times p_{li}^*) \\ &= \|n_{ijk}^*\| \frac{(n_{ijk}^*)^\top (p_{ji}^* \times p_{li}^*)}{\|n_{ijk}^*\|} = \|n_{ijk}^*\|. \end{aligned} \quad (48)$$

Notice from (11) that

$$\frac{n_{ijk}^*}{\|n_{ijk}^*\|} = \frac{p_{ki}^* \times p_{kj}^*}{\|p_{ki}^* \times p_{kj}^*\|} \quad (49)$$

for both $\{i, j, k\} \neq \{1, 2, 3\}$ and $\{i, j, k\} = \{1, 2, 3\}$. After applying (49) to (48), we obtain

$$\varphi_l^* = \frac{\|n_{ijk}^*\|}{\|p_{ki}^* \times p_{kj}^*\|} (p_{ki}^* \times p_{kj}^*)^\top (p_{ji}^* \times p_{li}^*). \quad (50)$$

From (11),

$$\frac{\|n_{ijk}^*\|}{\|p_{ki}^* \times p_{kj}^*\|} = \begin{cases} 1 & \text{if } \{i, j, k\} \neq \{1, 2, 3\} \\ \frac{1}{\|p_{31}^* \times p_{32}^*\|} & \text{if } \{i, j, k\} = \{1, 2, 3\}. \end{cases} \quad (51)$$

Given (18) and the fact that $\|p_{ji}^*\| = d_{ji}$, $\forall (j, i) \in \mathcal{E}$, it is obvious that (51) is only dependent on the desired distances. Now,

$$\begin{aligned} &= \frac{(p_{ki}^* \times p_{kj}^*)^\top (p_{ji}^* \times p_{li}^*)}{\|p_{ki}^* \times p_{kj}^*\| \|p_{ji}^* \times p_{li}^*\|} \\ &= \frac{(-p_{kj}^* \times p_{ki}^*)^\top (-p_{li}^* \times p_{ji}^*)}{\|p_{kj}^* \times p_{ki}^*\| \|p_{li}^* \times p_{ji}^*\|} \\ &= \frac{(-p_{kj}^* \times p_{ki}^* - p_{kj}^* \times p_{ji}^*)^\top (-p_{li}^* \times p_{ji}^* - p_{li}^* \times p_{ij}^*)}{\|p_{kj}^* \times p_{ki}^*\| \|p_{li}^* \times p_{ji}^*\|} \\ &= \|p_{kj}^*\| \|p_{li}^*\| \frac{\|p_{kj}^*\|^2 + \|p_{li}^*\|^2 - \|p_{ik}^*\|^2}{2 \|p_{kj}^*\| \|p_{li}^*\|} \|p_{ij}^*\|^2 \\ &\quad - \|p_{kj}^*\| \|p_{li}^*\| \frac{\|p_{kj}^*\|^2 + \|p_{li}^*\|^2 - \|p_{ik}^*\|^2}{2 \|p_{kj}^*\| \|p_{li}^*\|} \\ &\quad \cdot \|p_{ij}^*\| \|p_{ij}^*\| \frac{\|p_{ij}^*\|^2 + \|p_{ij}^*\|^2 - \|p_{ii}^*\|^2}{2 \|p_{ij}^*\| \|p_{ij}^*\|}, \end{aligned} \quad (52)$$

which is also only dependent on the desired distances. Thus, (50) is only a function of d_{ji} , $\forall (j, i) \in \mathcal{E}$.

Finally, it follows from (10c) that

$$\vartheta_l^* = (p_{li}^*)^\top n_{ijk}^* = (p_{li}^*)^\top n_{ijk}^* - (p_{li}^*)^\top n_{ijk}^* = (p_{li}^*)^\top n_{ijk}^*. \quad (53)$$

If $\{i, j, k\} \neq \{1, 2, 3\}$, then from (11)

$$\begin{aligned} \vartheta_l^* &= (p_{li}^*)^\top \frac{(p_{ki}^* \times p_{kj}^*)}{\|p_{ki}^* \times p_{kj}^*\|} \\ &= (p_{li}^*)^\top \frac{(p_{li}^* - p_{lk}^*) \times (p_{ij}^* - p_{ik}^*)}{\|p_{li}^* - p_{lk}^*\| \|p_{ij}^* - p_{ik}^*\|} \\ &= (p_{li}^*)^\top \frac{(p_{li}^* \times p_{ij}^* - p_{li}^* \times p_{ik}^* - p_{lk}^* \times p_{ij}^* + p_{lk}^* \times p_{ik}^*)}{\|p_{li}^* - p_{lk}^*\| \|p_{ij}^* - p_{ik}^*\|} \\ &= (p_{li}^*)^\top (p_{li}^* \times p_{ij}^*) = -6V_{ijkl}^* \end{aligned} \quad (54)$$

where $V_{ijkl}^* := V(p_i^*, p_j^*, p_k^*, p_l^*)$ from (2). Note that V_{ijkl}^* can be calculated using the Cayley-Menger determinant (Sommerville, 2011)

$$V_{ijkl}^* = \pm \frac{1}{288} \sqrt{\begin{vmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & \|p_{ji}^*\|^2 & \|p_{ki}^*\|^2 & \|p_{li}^*\|^2 \\ 1 & \|p_{ji}^*\|^2 & 0 & \|p_{kj}^*\|^2 & \|p_{ij}^*\|^2 \\ 1 & \|p_{ki}^*\|^2 & \|p_{kj}^*\|^2 & 0 & \|p_{ik}^*\|^2 \\ 1 & \|p_{li}^*\|^2 & \|p_{ij}^*\|^2 & \|p_{ik}^*\|^2 & 0 \end{vmatrix}}, \quad (55)$$

with $\|p_{ji}^*\| = d_{ji}$, where the sign is based on the desired ordering of vertices i, j, k per the convention described in Section 2.2. If $\{i, j, k\} = \{1, 2, 3\}$, then from (11) and the calculations in (54), we arrive at

$$\vartheta_l^* = -\frac{6V_{123l}^*}{\|p_{31}^* \times p_{32}^*\|} \quad (56)$$

where (18) is used to calculate the denominator in terms of the desired distances.

The desired projections ζ_2^* , ζ_3^* , and φ_3^* are special cases of the above variables. From (12), we have that

$$\zeta_2^* = (p_{21}^*)^\top \frac{p_{21}^*}{\|p_{21}^*\|} = \|p_{21}^*\| = d_{21}. \quad (57)$$

From (14) and (47), we obtain

$$\zeta_3^* = (p_{31}^*)^\top p_{21}^* = \frac{d_{31}^2 + d_{21}^2 - d_{32}^2}{2}.$$

Finally, from (15), we have

$$\begin{aligned} \varphi_3^* &= (p_{31}^*)^\top (n_{123}^* \times p_{21}^*) = (n_{123}^*)^\top (p_{31}^* \times p_{32}^*) \\ &= \|p_{31}^* \times p_{32}^*\| = 2S_{123}^* \end{aligned} \quad (58)$$

where $\tilde{S}_{123}^* := \tilde{S}_{123}(p_{31}^*, p_{32}^*)$ from (18) is only dependent on the desired distances.

References

- Anderson, B., Sun, Z., Sugie, T., Azuma, S., & Sakurama, K. (2017). Formation shape control with distance and area constraints. *IFAC Journal of Systems and Control*, 1, 2–12.
- Anderson, B., Yu, C., Fidan, B., & Hendrickx, J. M. (2008). Rigid graph control architectures for autonomous formations. *IEEE Control Systems*, 28(6), 48–63.
- Asimow, L., & Roth, B. (1979). The rigidity of graphs, II. *Journal of Mathematical Analysis and Applications*, 68(1), 171–190.
- Cao, Y., Sun, Z., Anderson, B., & Sugie, T. (2019). Almost global convergence for distance-and-area-constrained hierarchical formations without reflection. In *IEEE 15th International Conference on Control and Automation* (pp. 1534–1539).
- de Queiroz, M., Cai, X., & Feemster, M. (2019). *Formation Control of Multi-Agent Systems: A Graph Rigidity Approach*. Hoboken, NJ: Wiley.
- Eren, T., Whiteley, W., Anderson, B., Morse, A. S., & Belhumeur, P. N. (2005). Information structures to secure control of rigid formations with leader-follower architecture. In *Proceedings of the American Control Conference*, 2005. (pp. 2966–2971). IEEE.
- Ferreira-Vazquez, E. D., Flores-Godoy, J. J., Hernandez-Martinez, E. G., & Fernandez-Anaya, G. (2016). Adaptive control of distance-based spatial formations with planar and volume restrictions. In *2016 IEEE Conference on Control Applications* (pp. 905–910). Buenos Aires, Argentina.
- Ferreira-Vazquez, E. D., Hernandez-Martinez, E. G., Flores-Godoy, J. J., Fernandez-Anaya, G., & Paniagua-Contro, P. (2016). Distance-based formation control using angular information between robots. *Journal of Intelligent and Robotic Systems*, 83(3–4), 543–560.
- Grasgger, G., Koutschan, C., & Tsigaridas, E. (2018). Lower bounds on the number of realizations of rigid graphs. *Experimental Mathematics*, 6458, 1–12. <http://dx.doi.org/10.1080/10586458.2018.1437851>.
- Izmestiev, I. (2009). Infinitesimal rigidity of frameworks and surfaces. In *Lectures on Infinitesimal Rigidity*, Japan: Kyushu University.
- Jackson, B. (2007). Notes on the rigidity of graphs. In *Levico Conference Notes*, Vol. 4.
- Jing, G., & Wang, L. (2020). Multiagent flocking with angle-based formation shape control. *IEEE Transactions on Automatic Control*, 65(2), 817–823.
- Kang, S., Park, M., & Ahn, H. (2017). Distance-based cycle-free persistent formation: Global convergence and experimental test with a group of quadcopters. *IEEE Transactions on Industrial Electronics*, 64(1), 380–389.
- Khalil, H. K. (2015). *Nonlinear Control*. Pearson New York.
- Krick, L., Broucke, M. E., & Francis, B. A. (2009). Stabilisation of infinitesimally rigid formations of multi-robot networks. *International Journal of Control*, 82(3), 423–439.
- Lan, X., Xu, W., & Wei, Y. (2018). Adaptive 3D distance-based formation control of multiagent systems with unknown leader velocity and coplanar initial positions. *Complexity*, Article 1814653.
- Liu, T., Fernández-Kim, V., & de Queiroz, M. (2020). Switching formation shape control with distance+ area/angle feedback. *Systems & Control Letters*, 135, Article 104598.
- Liu, T., & de Queiroz, M. (2020). An orthogonal basis approach to formation shape control (extended version). [arXiv:2012.03064](https://arxiv.org/abs/2012.03064).
- Liu, T., & de Queiroz, M. (2021). Distance + angle-based control of 2D rigid formations. *IEEE Transactions on Cybernetics*, <http://dx.doi.org/10.1109/TCYB.2020.2973592>, in press.
- Liu, T., de Queiroz, M., & Sahebsara, F. (2020). Distance-based planar formation control using orthogonal variables. In *IEEE Conference on Control Technology and Applications* (pp. 64–69). Montreal, Canada.
- Liu, T., de Queiroz, M., Zhang, P., & Khaledyan, M. (2021). Further results on the distance and area control of planar formations. *International Journal of Control*, 94(3), 767–783.
- Mallison, H. V. (1935). The use of signs in geometry. *The Mathematical Gazette*, 19(233), 124–130. URL: <http://www.jstor.org/stable/3608027>.
- Mou, S., Cao, M., & Morse, A. S. (2015). Target-point formation control. *Automatica*, 61, 113–118.
- Oh, K., Park, M., & Ahn, H. (2015). A survey of multi-agent formation control. *Automatica*, 53, 424–440.
- Sommerville, D. (2011). *An Introduction to the Geometry of N Dimensions*. New Academic Science.
- Sugie, T., Anderson, B., Sun, Z., & Dong, H. (2018). On a hierarchical control strategy for multi-agent formation without reflection. In *2018 IEEE Conference on Decision and Control* (pp. 2023–2028).
- Summers, T. H., Yu, C., Dasgupta, S., & Anderson, B. (2011). Control of minimally persistent leader-remote-follower and coleader formations in the plane. *IEEE Transactions on Automatic Control*, 56(12), 2778–2792.
- Yu, C., Hendrickx, J. M., Fidan, B., Anderson, B., & Blondel, V. D. (2007). Three and higher dimensional autonomous formations: Rigidity, persistence and structural persistence. *Automatica*, 43(3), 387–402.
- Zwillinger, D. (2002). *CRC Standard Mathematical Tables and Formulae* (31st ed.). CRC Press.



Tairan Liu received the B.Sc. degree in Theoretical and Applied Mechanics from the University of Science and Technology of China, Hefei, Anhui, P.R. China, in 2012. He received the Ph.D. degree in Mechanical Engineering in 2020 from Louisiana State University, Baton Rouge, Louisiana, USA. He is currently a Postdoctoral Fellow with the University of Georgia, Athens, Georgia, USA. His research interests include nonlinear control, multiagent systems, and robotics.



Marcio de Queiroz received a B.S. Degree in Electrical Engineering from the Federal University of Rio de Janeiro, Brazil, a M.S. Degree in Mechanical Engineering from the Pontifical Catholic University of Rio de Janeiro, Brazil, and a Ph.D. Degree in Electrical Engineering from Clemson University, Clemson, SC in 1997. From August 1997 to August 1998, he was a Post-Doctoral Researcher at the Rotating Machinery and Controls Laboratories of the University of Virginia. From September 1998 to May 2000, he was a Visiting Assistant Professor at the Department of Mechanical and Aerospace Engineering of the NYU Tandon School of Engineering. In July 2000, he joined the Department of Mechanical and Industrial Engineering at LSU, where he is currently the Roy O. Martin Lumber Company Professor. Dr. Queiroz served as an Associate Editor for the IEEE Transactions on Automatic Control, the ASME Journal of Dynamic Systems, Measurement, and Control, the IEEE/ASME Transactions on Mechatronics, and the IEEE Transactions on Systems, Man, and Cybernetics – Part B. His research interests include nonlinear control theory and applications, multiagent systems, robotics, active magnetic and mechanical bearings, and biological/biomedical system modeling and control. Dr. Queiroz is a Fellow of the American Society of Mechanical Engineers (ASME).